C++ in an AI World

JODY HAGINS
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Lock-Free Timer Scheduling With C++ Atomics

MAXIM GURSCHI

Jody Hagins
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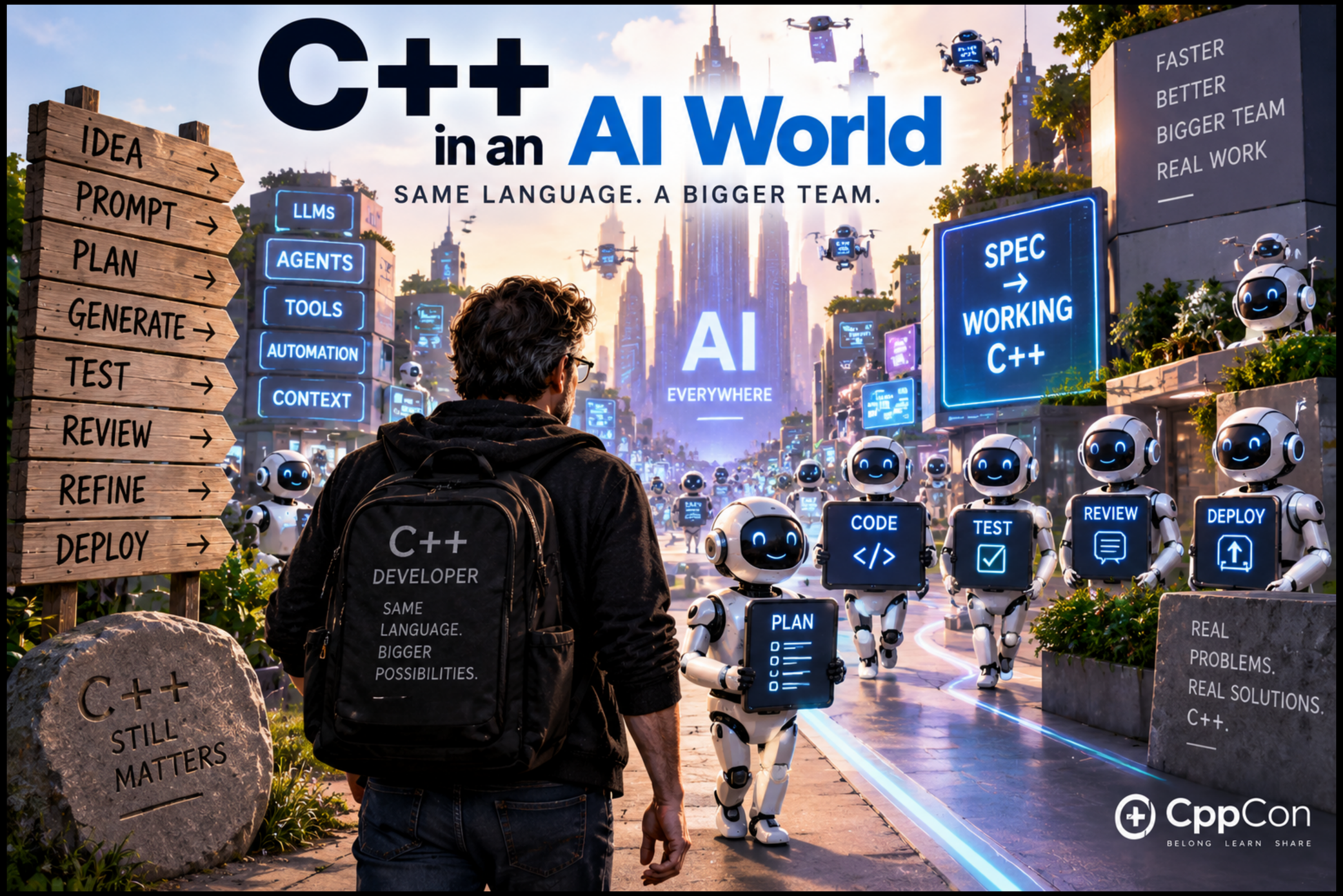

C++ in an AI World
SAME LANGUAGE. A BIGGER TEAM.
IDEA
PROMPT
PLAN
GENERATE
TEST
REVIEW
REFINE
DEPLOY
LLMs
AGENTS
TOOLS
AUTOMATION
CONTEXT
AI
EVERYWHERE
SPEC → WORKING C++
FASTER
BETTER
BIGGER TEAM
REAL WORK
C++
DEVELOPER
SAME LANGUAGE. BIGGER POSSIBILITIES.
C++ STILL MATTERS
PLAN
CODE
TEST
REVIEW
DEPLOY
REAL PROBLEMS. REAL SOLUTIONS. C++.
CppCon
BELONG LEARN SHARE

AI in a C++ World
AI GETTING ITS BUTT KICKED IN A C++ WORLD
Templates ✓
Move Semantics ✓
Lifetimes ✓
Undefined Behavior ✓
The Standard ✓
Compile Errors ✓
Link Errors ✓
Reality ✓
C++ ALWAYS WINS (HUMAN, TOO)
C++
REAL WORLD
C++
DOESN'T COMPILE
TEMPLATES
AI
LIFETIMES
WRONG TYPE
BROKEN LOGIC
UB
HALLUCINATION
LAST YEAR...

C++ in an AI World
Same Language.
A Bigger Team.
Real Progress.
THE WORKFLOW
IDEAS & SPECS
PLAN (AGENTS)
GENERATE
BUILD & TEST
REVIEW & ITERATE
PRODUCTION
REAL VALUE
while (ideas) {
auto plan = ai.plan();
auto code = ai.generate();
auto result = build_and_test(code);
improve(result);
}
// Real progress.
AI
C++
GOOD CODE TOGETHER
PROMPTS THAT WORK
AGENTIC WORKFLOWS
REAL C++ PROJECTS
TESTS. ALWAYS.
PRODUCTION LESSONS
Better Tools
Better Code
Bigger Impact
HUMAN JUDGMENT STILL MATTERS
Same Language. Bigger Possibilities.
THIS YEAR...

The Power of Knowing What You Don't Know

THINK AGAIN

ADAM GRANT

#1 *New York Times* bestselling author of
ORIGINALS

C++ now

2026

AI is Forcing us to Obey Scott Meyers

Jody Hagins

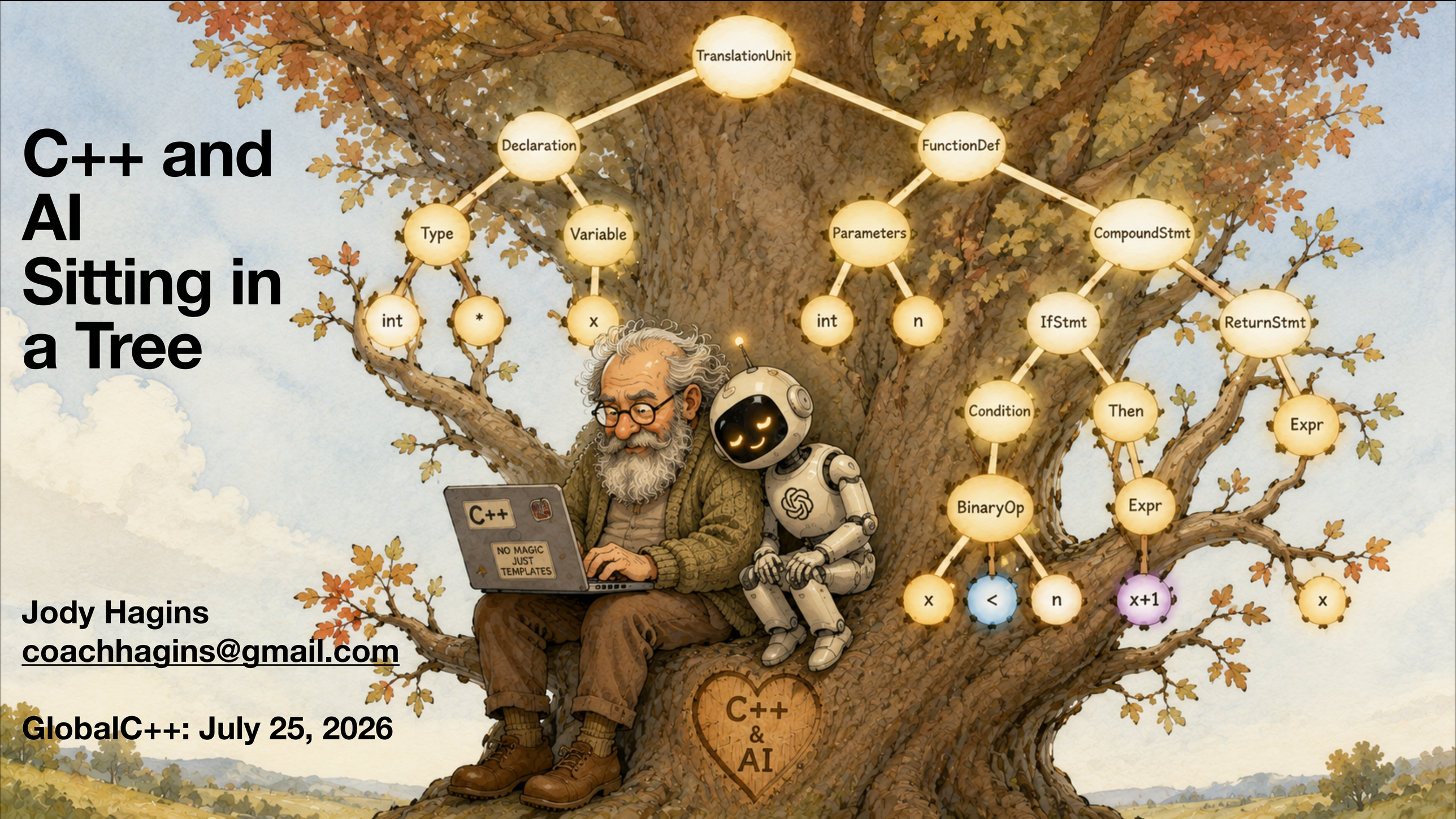
C++ now

2026

What Every C++ Programmer Needs to Know About AI

Jody Hagins

C++ and AI Sitting in a Tree



Jody Hagins
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GlobalC++: July 25, 2026

First Things First

Ask questions

First Things First

Ask questions

I'm not an expert

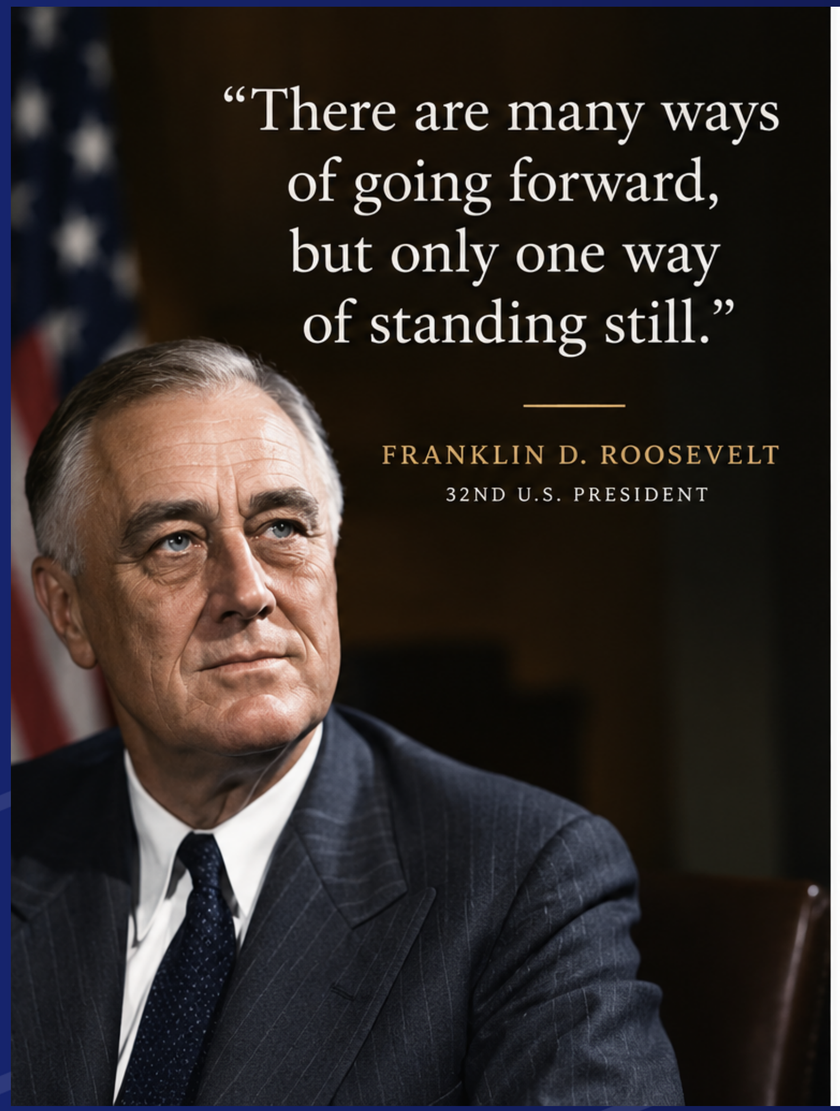
First Things First

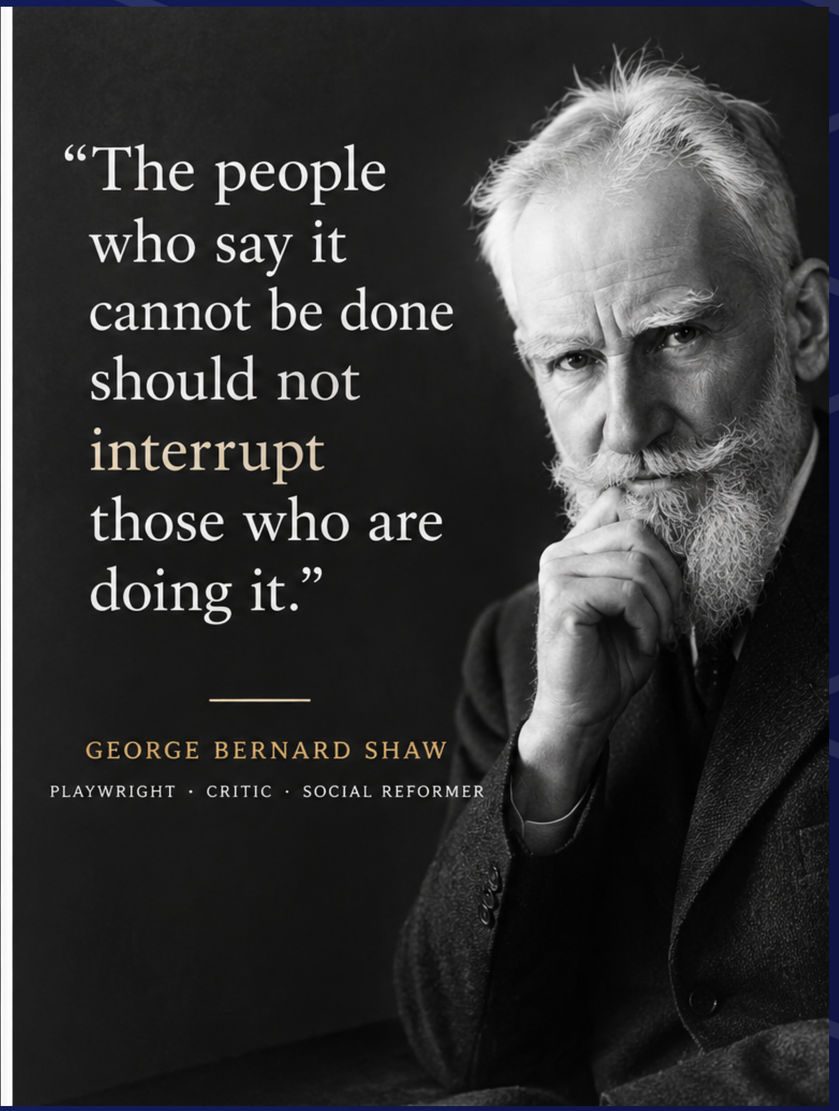
Ask questions

I'm not an expert

There are no experts yet

First Things First





Talk Breakdown

Basics

My Journey

What I'm doing today (and why)

AI

Generative AI - tokens generated by LLMs

Publications You Will Never Read

Hagins, Biswas, Yu, Model-Based Diagnosis in the Process-Ontology Framework, The Second AAAI Workshop on Model Based Reasoning, Boston, MA, July 1990

Biswas, Yu, Hagins, Strobel, Kendall, Cannon, Bezdek, An Efficient Scheme for Characterizing Hydrocarbon Plays for Analogical Analysis, AAPG Annual Convention, San Francisco, CA, June 1990

Biswas, Yu, Hagins, Strobel, Kendall, Cannon, Bezdek, PLAYMAKER: A Knowledge-Based Approach to Characterizing Hydrocarbon Plays, Applications of AI VIII (SPIE), Orlando, FL, April 1990

Biswas, Strobel, Hagins, Kendall, Cannon, Bezdek, An Associational Scheme for Characterizing Hydrocarbon Plays for Analogical Reasoning, IEEE Expert, March 1990

Biswas, Hagins, Debelak, Qualitative Modeling in Engineering Applications, 1989 IEEE Conference on Systems, Man, and Cybernetics, Cambridge, MA, November 1989

Weinberg, Hagins, Biswas, Extending Temporal Reasoning in Process-Oriented Qualitative Reasoning, Proceedings of IJCAI-89 Workshop on Model Based Reasoning, Detroit, MI, August 1989

Debelak, Biswas, Hagins, Qualitative Modeling in Chemical Engineering Applications, American Institute of Chemical Engineers: 1989 Summer National Meeting, August 20, 1989

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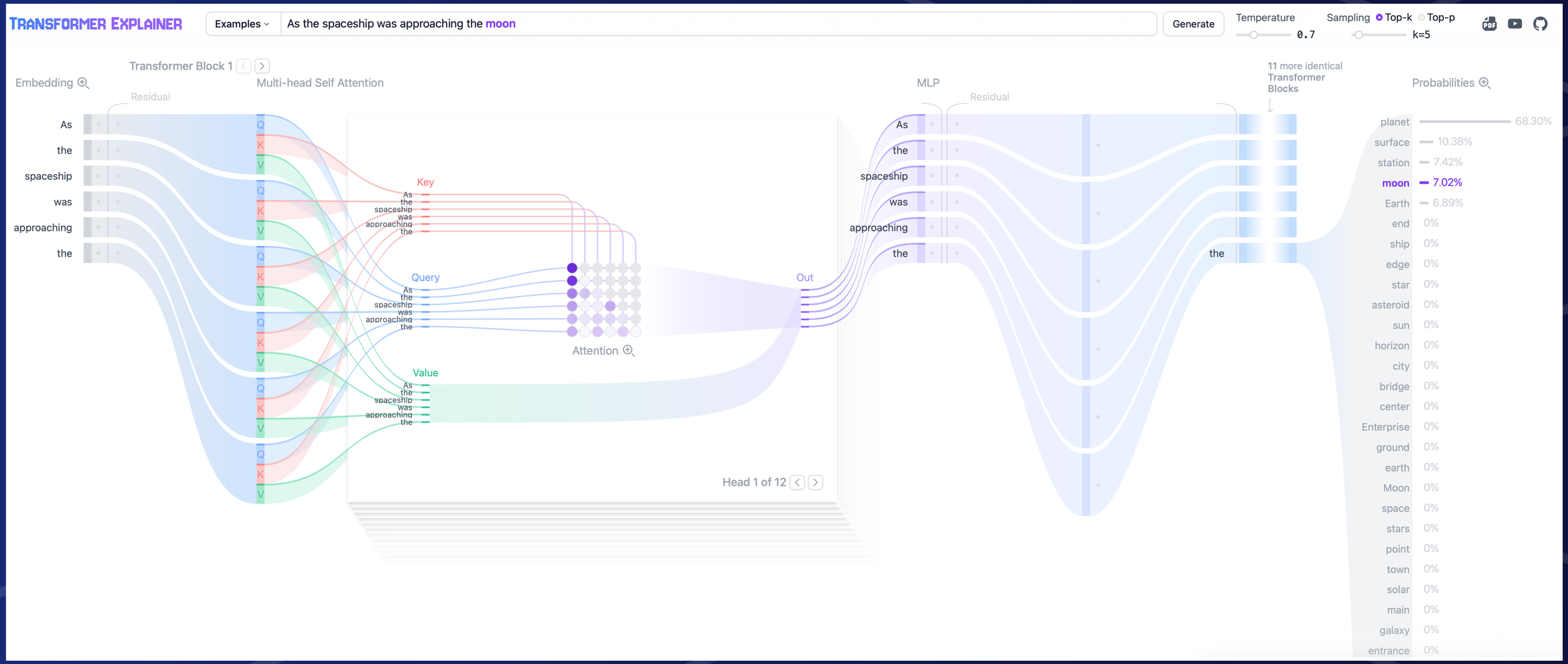
AI

Generative AI - text generated by LLMs

https://poloclub.github.io/transformer-explainer/

AI



https://poloclub.github.io/transformer-explainer/

AI

LLMs are like a box of chocolates
You never know what you're going to get.
Same prompt. Different chocolate.
BRILLIANT INSIGHT
ALMOST RIGHT
COMPLETELY WRONG
GENERIC PLATITUDE
SURPRISINGLY CREATIVE
OUTDATED INFORMATION
GREAT IDEA
HALLUCINATION
VERBOSE RAMBLING
EXACTLY WHAT I NEEDED
SAFE AND BORING
REFUSAL
PROMISING BUT INCOMPLETE
TOTALLY OFF TOPIC
A PLEASANT SURPRISE
SAME PROMPT. DIFFERENT OUTCOME.

AI

Generative AI - text generated by LLMs

https://alonsosilva-perplexity.hf.space/

AI

If I start counting to ten...

one two three four five six seven eight nine ten

... you can see that as the sequence progresses, the LM gets better at predicting the sequence and by the end, it's 100% correct about how things are going to continue.

The first time I write this sentence, the model is quite confused about what token is about to come next, especially if I throw in weird words like pumpkin, clown, tweets, alpha, teddy bear.

But the second time? It expects almost every word that appears:

The first time I write this sentence, the model is quite confused about what token is about to come next, especially if I throw in weird words like pumpkin, clown, tweets, alpha, teddy bear.

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↵

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↵

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↵

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↵

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Can I Trust LLMs?

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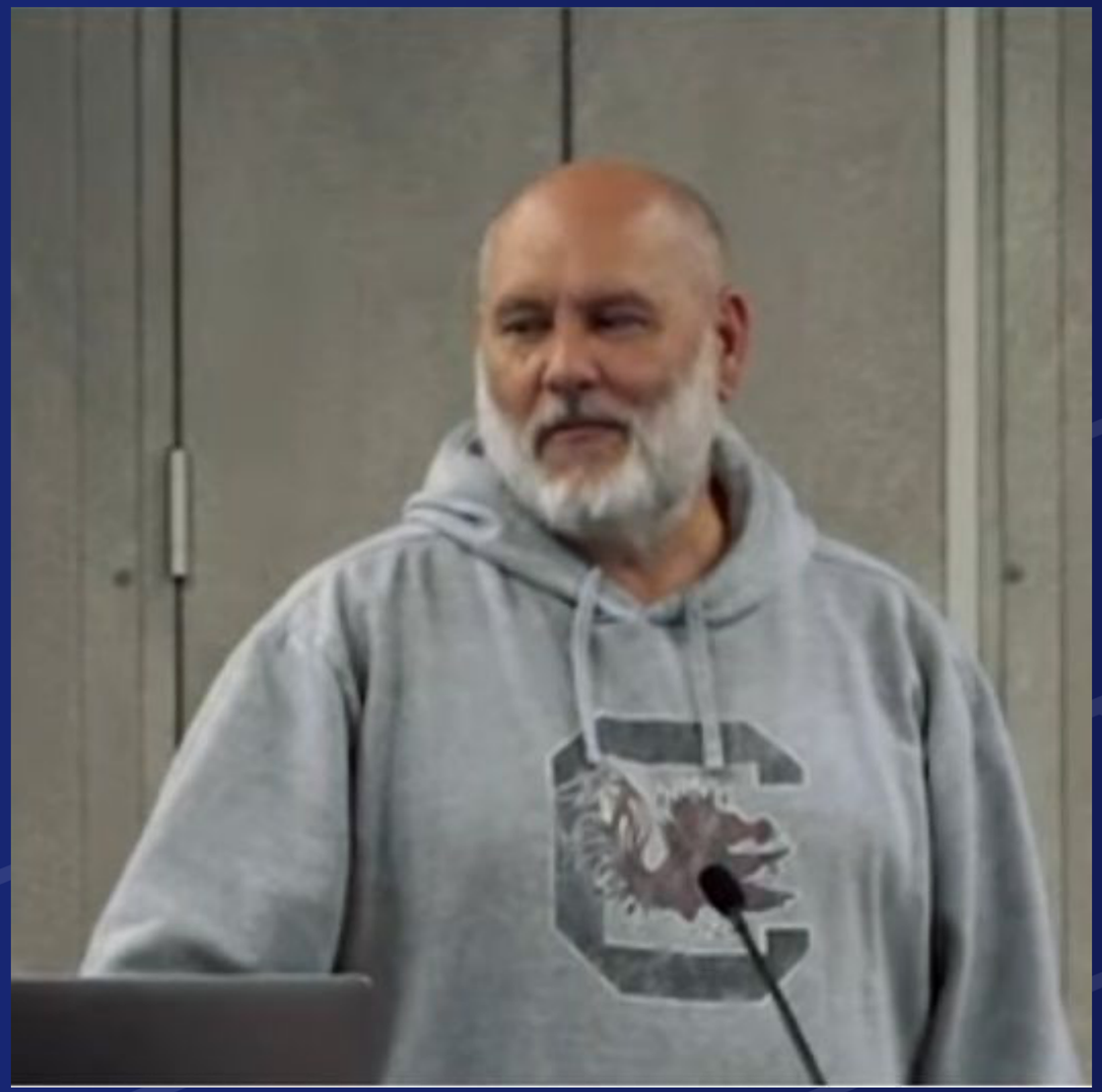

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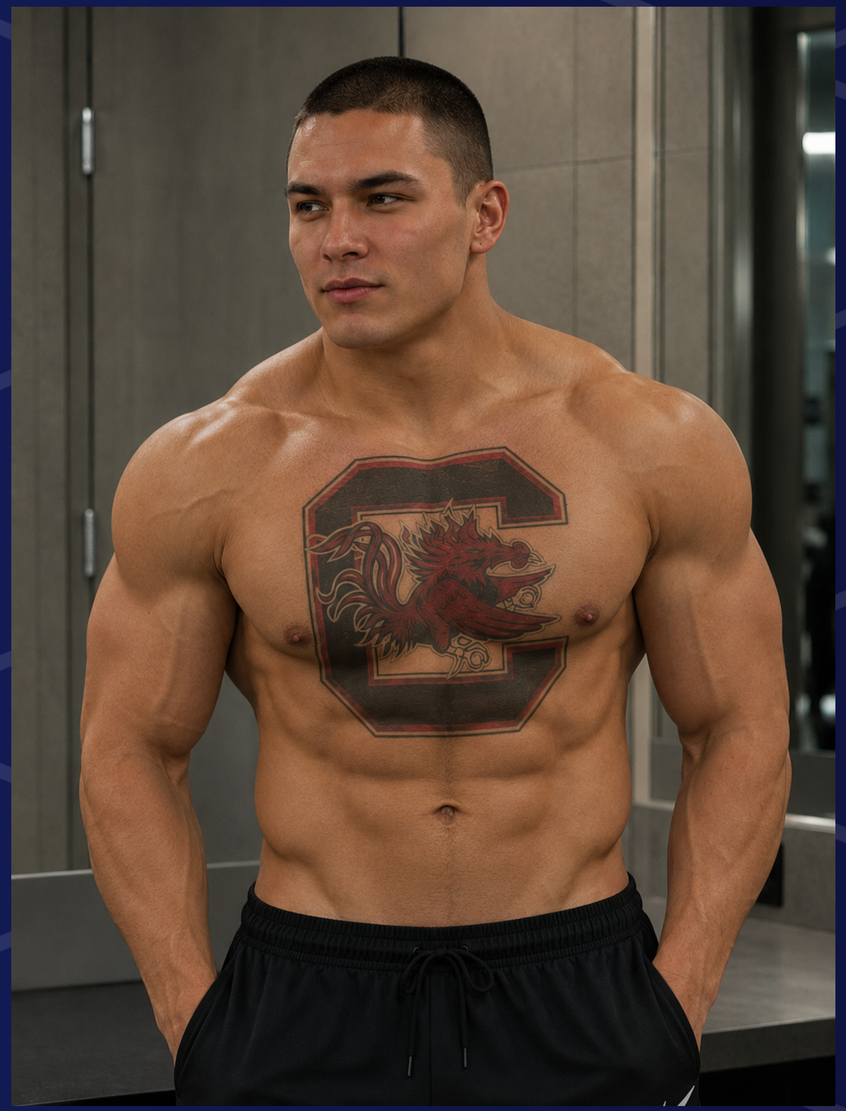

Can I Trust LLMs?

C++ is H-A-R-D

Many argue C++ is an expert-only language

Four Coding Harness Pillars

Models

What they are
How they generate text

Four Coding Harness Pillars

Models

What they are
How they generate text

Prompts

How you communicate intent to a model

Four Coding Harness Pillars

Models

What they are
How they generate text

Prompts

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Context

How the model "remembers" (it doesn’t)

Four Coding Harness Pillars

Models

What they are
How they generate text

Prompts

How you communicate intent to a model

Context

How the model "remembers" (it doesn’t)

Tools

How models interact with the world (you do)

Three Goals of Agentic Coding

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Manage Context

All context: prompts, tools, output, shared, size, content, etc...

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Reduce Non-determinism

Extract as much determinism as possible - leave only model things for the model

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Maximize Observability

Log everything, reap logs for both human and AI learning and automation

Three Goals of Agentic Coding

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Reduce Non-determinism

Extract as much determinism as possible - leave only model things for the model

Maximize Observability

Log everything, reap logs for both human and AI learning and automation

Automation

Most things are autonomous.
HITL is the exception

My Journey

2023: ChatGPT
2023: OpenRouter.ai
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2023: ChatGPT

Litmus Test: Build complete feed handler from specs

2023: ChatGPT

1.4.3 Order Cancel Message

This message is sent whenever an order on the book is modified as a result of a partial cancellation.

Order Cancel Message

Name	Offset	Length	Value	Notes
Message Type	0	1	"X"	Order Cancel Message
Stock Locate	1	2	Integer	Locate code identifying the security
Tracking Number	3	2	Integer	Nasdaq internal tracking number
Timestamp	5	6	Integer	Nanoseconds since midnight
Order Reference Number	11	8	Integer	The reference number of the order being canceled
Cancelled Shares	19	4	Integer	The number of shares being removed from the display size of the order as a result of a cancellation

2023: ChatGPT

Litmus Test: Build complete feed handler from specs

Indistinguishable from my own

2023: ChatGPT

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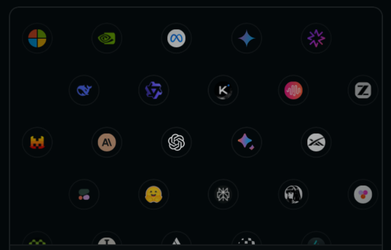
2023: OpenRouter.ai

The Unified Interface For Every Model

Better prices, better uptime, no subscriptions.

Get API Key | Discover Models

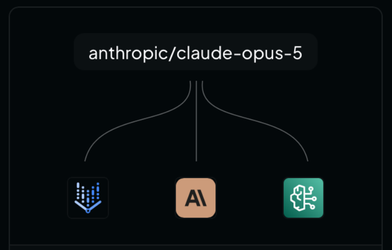
400T+	10M+	80+	500+
Monthly Tokens	Global Users	Providers	Models



Text, Images, Videos, and Audio

Generate anything through a single, unified interface. All major models in one place.

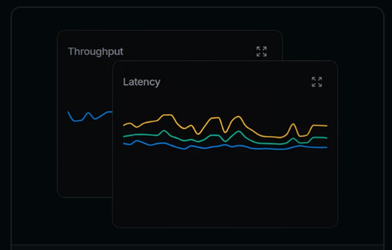
Browse all



Higher Availability

Reliable AI models via our distributed infrastructure. Fall back to other providers when one goes down.

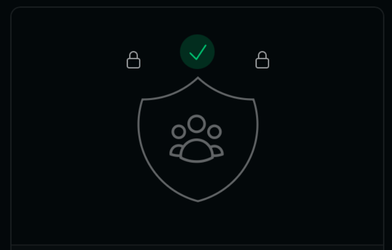
Learn more



Price and Performance

Keep costs in check without sacrificing speed. OpenRouter runs at the edge for minimal latency between your users and their inference.

Learn more



Custom Data Policies

Protect your organization with fine grained data policies. Ensure prompts only go to the models and providers you trust.

View docs

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2024: Sonnet 3.5

- First time to process complete PDF
- No more python to split PDFs
- No more text file with page numbers for all tables
- Only Sonnet models for a long time

2024: Sonnet 3.5

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2024: Sonnet 3.7

- First time to go directly from spec to code
- Only for representation
- I was getting better at managing context
- Code review personas

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2024: Sonnet 4

- Go from spec to code plus dispatching
- Tests were MUCH better
- Code Review was really, really good
- I had gotten really good at managing context

2024: Sonnet 4

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2025: CppCon

- A number of talks that mentioned AI
- I asked the same question

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Sep 29, 2025: Sonnet 4.5

- Amazing
- Me or the model?

Sep 29, 2025: Sonnet 4.5

- Amazing
- Me or the model?
- Much less 💩 but still 💩

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Oct 9, 2025: OPRA

- Options feed handler, fairly simple
- Complete implementation with tests
- Including unique A/B arbitrage algorithm
- Not production quality, but usable as a baseline
- Hooks trump rules
- Still some 💩

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Nov 24, 2025: Opus 4.5

- My quality had been improving as I improved my process
- I had come up with ways to make Sonnet better
- Opus was still a significant jump
- Code review
- Test spec generation
- Not much 💩

Nov 24, 2025: Opus 4.5

- My quality had been improving as I improved my process
- I had come up with ways to make Sonnet better
- Opus was still a significant jump
- Code review
- Test spec generation
- Not much 💩
- I've not written a line of C++ code since November*

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Dec, 2025: Production FH

- Implemented production quality feed handler
- I didn't write a line of code
- Tests were praised as example to be imitated
- Indistinguishable from my own code
- I was happy to commit and take responsibility
- No 💩

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Dec, 2025: Orchestrator

- Started building a full orchestrator
- Automate the process I had been doing manually
- Lots of hooks and scripts, specific to claude code
- Ant Farm is still a pipe dream, but there is some hope

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- Continuously running invocation
- Fully tested
- FIX Client
- FIX Server
- Sequencer
- Matching Engine
- Ant Farm is no longer a pipe dream - a matter of when

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Mar 4, 2026: Maverick

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http://www.incompleteideas.net/IncIdeas/BitterLesson.html

Mar 4, 2026: Maverick

- The graph idea is based on the StrongDB Attractor
  - https://github.com/strongdm/attractor
- Mine has lots of extra features for C++ and my preferences
- Gets very close to AntFarm: Queue of Tasks

Mar 4, 2026: Maverick

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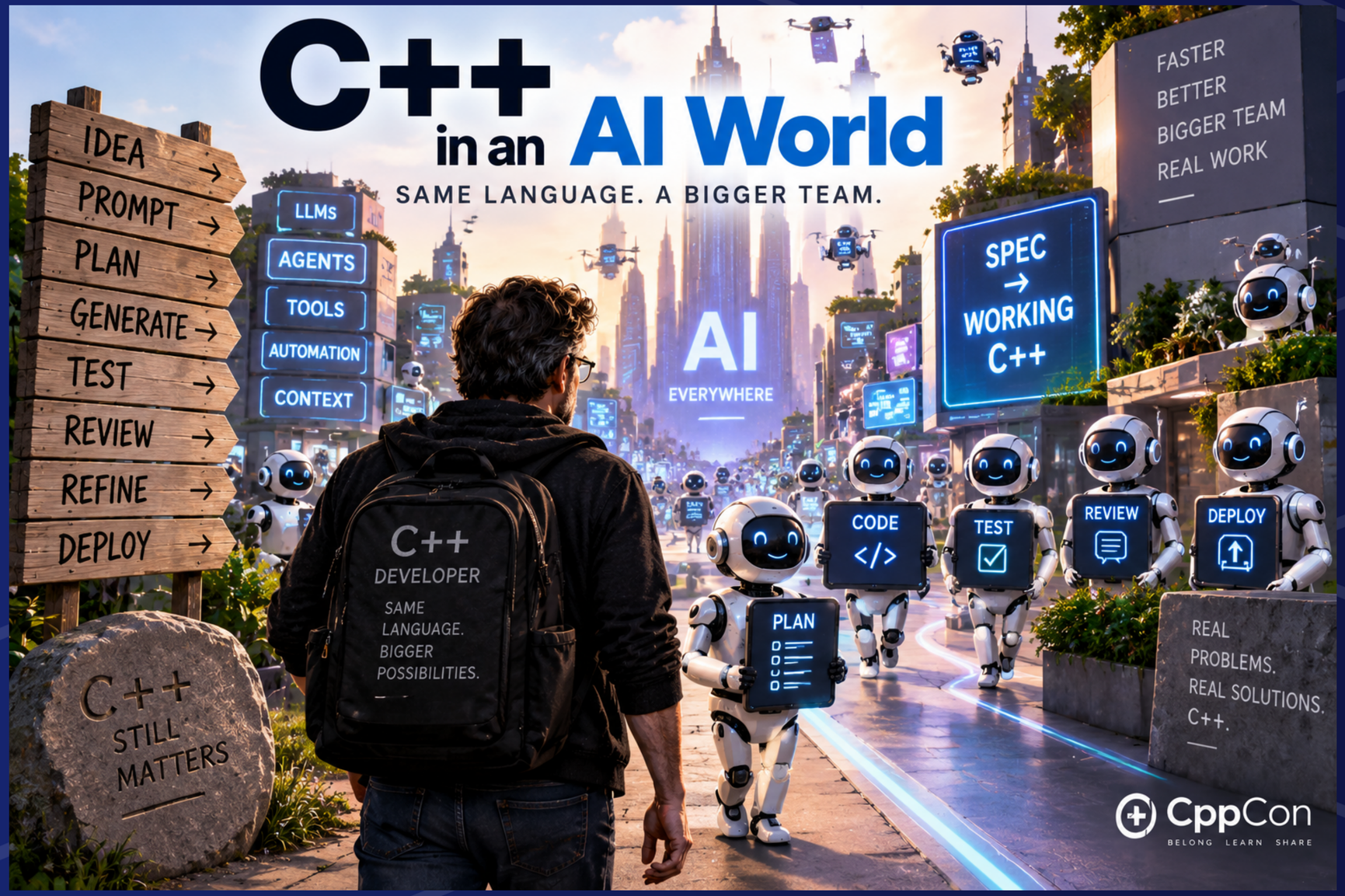
Q & A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